Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **AI-enabled SaaS**, **TypeScript/React applications**, and reliable **Python/SQL services** with automated testing, CI/CD, monitoring, and human-validated workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant FastAPI orchestration** with explicit job states, background workers, failure isolation, recoverable processing, and observable production workflows under concurrent load.
- Achieved 99%+ production reliability across 5K+ monthly jobs through **data validation**, reconciliation checks, error handling, cloud storage integration, and CI/CD releases connecting FastAPI services and relational databases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround 60% by delivering a **React and Flask platform** for 150+ instructors and 10K+ daily grading requests, integrating multimodal LLMs, OCR, REST APIs, and AWS RDS.
- Maintained 90%+ grading consistency across 10K+ submissions by integrating **GPT-4o, Claude, Qwen-VL, and OCR**, rubric generation, model switching, fallback paths, and persistent scoring workflows with instructor review controls.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **real-time microservice platform** using React, TypeScript, Node.js, WebSockets, REST APIs, and Azure SQL across distributed business workflows and customer-facing operational requirements.
- Reduced resource wastage by 42% through **testing and production monitoring**, integrating automated suites, Docker/Jenkins delivery, Azure Monitor, Sentry, and Kibana across modular production services reliably at scale.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **Flask APIs** for a cross-platform service-booking product covering time slots, pricing, coupons, payments, refunds, disputes, and real-time Firebase synchronization across 50K+ booking records with responsive workflows.
- Reduced manual refund handling approximately 70% by integrating **payment webhooks**, status updates, dispute workflows, and sub-200ms booking APIs across mobile and web experiences for operational teams reliably.

Skills

Languages: TypeScript, React, Python, SQL, SQL Server, JavaScript, Java, C++
Backend & Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, Azure SQL, AWS RDS, MongoDB, Redis, Data Modeling
AI & Product: LLMs, GPT-4o, Claude, Qwen-VL, OCR, Document Retrieval, Semantic Search, Prompt Engineering, Human Review Workflows
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Git, Cloud Monitoring
Engineering: Automated Testing, Unit Testing, Integration Testing, PyTest, Mockito, API Design, Microservices, System Design, Performance Tuning, Production Troubleshooting

Projects

AI-Assisted EMS Documentation Validator - Design Concept

- Designed an **AI-assisted EMS documentation validator** concept using Python, React, SQL Server, configurable rules, human review checkpoints, automated tests, audit trails, and production monitoring for secure clinical workflows.

HMIS - Hospital Management Information System | Live

- Built a **TypeScript healthcare platform** with ABDM APIs, consent-based patient-history access, appointment workflows, Twilio alerts, and Sentry monitoring, reducing administrative workload 30% while supporting 100+ daily appointments.

Big-Brain - AI Document Platform | Live

- Built a secure **AI document platform** over 50K+ documents using LLaMA-2, FAISS, Pinecone, PostgreSQL, and REST APIs, enabling contextual Q&A, summarization, and sub-second retrieval workflows reliably and securely.